

Q1:

A 12 TB dataset is stored in HDFS with:

- Block size = 256 MB
- Replication factor = 3
- Cluster has 40 DataNodes, each with 10 TB usable storage

- a) Calculate total number of blocks.
- b) Total storage consumed including replication.
- c) Average blocks per DataNode (assume uniform distribution).
- d) If one node fails, how many blocks need re-replication?
- e) Explain how rack awareness affects this calculation.

Q2:

Dataset size: 1 billion points

K = 50

Dimensions = 200

Iterations = 20

- a) Derive time complexity.
- b) Estimate total distance computations.
- c) Suggest distributed optimization strategy in Spark.
- d) Compare with Mini-Batch K-Means in terms of bias-variance tradeoff.

Q5:

Suppose rule found:

{Milk, Bread} → {Butter}

Support = 0.6%

Confidence = 85%

Is this rule strong? Justify using lift. Can high confidence still be misleading? If Butter is purchased in 80% transactions, what is lift? Should this rule be used for recommendation? Why or why not?

Q6:

A healthcare analytics system collects 2 TB of patient data daily, including MRI images, text prescriptions, and real-time sensor data from wearable devices.

- a) Identify which Big Data characteristics are represented in this scenario.
- b) Explain how Velocity and Variety together create challenges for data processing systems.
- c) Suggest a suitable Big Data framework to manage this system and justify.

Q7:

Consider a Hadoop cluster with 10 nodes, each having:

- 16 GB RAM
- 8 CPU cores

Each container requires 2 GB RAM and 1 CPU core.

- a) Calculate the maximum number of containers per node.
- b) Determine the total containers available in the cluster.
- c) Explain how YARN improves scalability compared to Hadoop 1.0 architecture.

A Hadoop cluster has:

- 1 NameNode
- 1 Standby NameNode
- 8 DataNodes
- Block size = 128 MB
- Replication factor = 3

A 1 TB file is uploaded to HDFS.

Calculate:

Total number of blocks created. Total physical storage consumed in the cluster.

If 2 DataNodes fail simultaneously:

What percentage of blocks will still remain available (assume ideal uniform block distribution)? Explain how HDFS maintains fault tolerance in this case.

In YARN:

A MapReduce job requires 120 containers, each with 4GB RAM and 2 vCores.

Each node has 64GB RAM and 16 vCores. What is the maximum number of containers per node? Minimum number of nodes required? Explain how the ResourceManager and NodeManager coordinate container allocation.

Q3:

A digital payment company processes:

- 1.5 billion transactions/month
- 40 million users
- Fraud rate = 0.8%
- Daily data generation = 25 TB
- Cluster: 300 nodes
- Each node: 256 GB RAM, 64 cores

Requirements:

- Fraud detection in < 2 seconds
 - Monthly customer segmentation
 - Purchase prediction
 - Must maintain 99.99% availability
- As Lead Data Scientist, answer the following:
- A. HDFS & Cluster Planning
 - Estimate monthly storage requirement including replication factor = 3.
 - If block size increases from 128 MB to 512 MB:
 - How does it affect parallelism?
 - NameNode memory usage?
 - Fault recovery time?
 - B. Fraud Detection using Naïve Bayes
 - Fraud rate = 0.8%.
 - Explain mathematically why accuracy is a misleading metric.
 - Suggest better evaluation metrics.

If two features are highly correlated (correlation = 0.9), explain how Naïve Bayes probability estimation is distorted.

Propose alternative models and justify trade-offs.

C. Customer Segmentation using K-Means 40 million users, 100 features, $K = 100$.

- Estimate number of distance computations per iteration.
- Suggest dimensionality reduction approach.
- Compare Mini-Batch K-Means vs full K-Means.
- D. Decision Tree at Scale
- Tree depth = 18 with 500,000 leaf nodes.
- Analyze bias-variance tradeoff.
- Explain cost-complexity pruning mathematically.

Would Random Forest improve fraud detection stability? Justify with ensemble theory.

Q8:

A dataset contains 20 samples:

- 12 Positive
- 8 Negative

Entropy: $\text{Entropy}(S) = -p_1 \log_2 p_1 - p_2 \log_2 p_2$

Attribute A splits data into:

Subset 1: 8 Positive, 2 Negative

Subset 2: 4 Positive, 6 Negative

- Compute Entropy(S).
- Compute Entropy after split.
- Calculate Information Gain for attribute A.
- What is overfitting in decision trees?
- How do pruning techniques reduce overfitting?
- Compare pre-pruning vs post-pruning in Big Data datasets.